

Optimal Design Methods to Transform 3D NAND Flash into a High-Density, High-Bandwidth and Low-Power Nonvolatile Computing in Memory (nvCIM) Accelerator for Deep-Learning Neural Networks (DNN)

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Abstract: We propose optimal design methods of 3D NAND Flash to achieve high-density, high-bandwidth and low-power nvCIM. By suitably engineering the device, we can produce ultra-low ON current of 2nA (mean) at saturated region instead of subthreshold region, while the OFF leakage current is much below 1pA. Such low Ion and large ON/OFF ratio provide large bandwidth to parallelly sum more than 10'000 cells together to offer high efficiency for DNN computing. The three-dimensional summation in 3D NAND also allows effective multi-bit resolution of weight without resorting to complex analog memory design. For the first time we witnessed the power of “central limit theory” in 3D NAND nvCIM, where the large number of summation averages out the noise and provides high accuracy of MAC. The effect of non-ideal cell variations, noises and shifts are studied systematically. Through adequate calibration techniques the 3D NAND nvCIM can provide accuracy close to the software limitation, with reasonable tolerance to various device errors. The 3D NAND nvCIM is promising to be an energy-efficient (TOPS/W~40) edge computing solution for large neural networks (>100Mb weight).

I. Introduction

Computing in memory (CIM) [1-4] has got great attention recently as a potential low-power solution of “non-Von-Neumann” architecture to realize efficient computing for the deep learning neural network (DNN). **Figure 1(a)** illustrates a typical DNN procedure for image recognition. It consists of many convolution layers that involve extensive multiple-and-accumulate (MAC) operations, as shown in **Fig. 1(b)**. The MAC requires extensive vector matrix multiplication of weights (stored in memory) and inputs, and is very power consuming in large data movements. CIM can drastically save the data movements of MAC.

The concept of CIM is quite simple. Usually it is realized by using memory cell's conductance (I_{cell}/V_{BL}) to represent the weights. By summing currents of plural memory cells connected to a sense amplifier (SA), the sum corresponds to the MAC value. It is often demonstrated by resistive memory devices (ReRAM) [2-3] or NOR-type Flash memory [5]. For resistive memory devices, the performance bottleneck is often gated by the poor ON/OFF ratio of ReRAM that cannot allow summing many cells. Usually the I_{ON} needs to be at least a few μA for ReRAM to keep sufficient ON/OFF ratio, which means that to sum 100 cells the total current already exceeds 100 μA . It is too large for the SA design and is very power consuming. Another attempt is to use NOR Flash at subthreshold region to produce low I_{ON} . However, the subthreshold sensing means that V_g is close to V_t . A minor V_t shift (e.g. 50mV) caused by cell noise, variation or aging will drastically impact I_{cell} in the subthreshold region, leading to poor accuracy.

In this work, we propose using 3D NAND Flash device to enable high-bandwidth summation with high accuracy and good tolerance to device error. Different from the “analog design” [5] of CIM that produces multi-bit computing, our strategy is to use single-level cell (SLC) to provide high reliability and error tolerance. Because 3D NAND has an invincible high-density advantage over other memory devices, we can duplicate the memory cells many times to produce effective multi-bit resolutions, still providing very low cost and low-power solution.

II. Detail Design Methods and Characteristics:

(a) Methods to produce MAC in 3D NAND

Figure 2 illustrates the method of producing MAC. We select a

WL each time, and sum the string current from many BL's to collect the summed current at the source line. The source line of each block is separated and connected to a corresponding SA, as shown in **Fig. 3(a)**. Note that in 3D NAND we have an additional terminal of string select transistor (SSL). We can also turn-on many SSL's together to allow an effective 3D summation:

$$I_{SL}(j) = w(1-1,j) * V_{BL1} + w(2-1,j) * V_{BL2} + \dots \dots \dots (\text{in SSL1}) \\ + w(1-2,j) * V_{BL1} + w(2-2,j) * V_{BL2} + \dots \dots \dots (\text{in SSL2}) \\ + w(1-N,j) * V_{BL1} + w(2-N,j) * V_{BL2} + \dots \dots \dots (\text{in SSLN})$$

$w(i,j)$ is the conductance corresponding to a weight. The many BL's in NAND serve as the inputs. We suggest SLC (for weight and input) for the high reliability and easy design without resorting to complex digital to analog converter. On the other hand, the many SSL's provides a means to distribute the weight in multiple cells. For example, a 3-SSL design means that the weight value can represent 0, 1, 2, 3 levels (2bit) with SLC storage for each cell. It can be generalized to various scenarios of multi bits.

Figure 3(b) shows the advantage of nvCIM. The 3D NAND serves as the “MAC accelerator” so that the system can greatly reduce the SRAM and ALU, and minimize data transfer through bus. **Figure 4** illustrates the overall schematic diagram of 3D NAND nvCIM. The many blocks correspond to the M kernels of DNN. Each block's source line connects to a SA, which needs to produce multi-bit resolution for MAC. Array is designed with a small tile to reduce RC delay.

The 3D NAND nvCIM needs two types of sense amplifier at X and Y directions. The original page buffer (PB) is needed to provide ordinary array operation for programming the device. The neural network is remotely trained and then downloaded on the chip by programming each cell via PB. The ordinary read speed may be slow to detect each bit via PB. On the other hand, the summed MAC current at source line will be in the range of $\sim 10\mu A$, which enable much faster SA speed (Tread $\sim 1\mu sec$ with multi-bit output) for MAC computing.

Different from the previous proposal [6] where inputs are at WL's and outputs at BL's, our proposal makes BL's as input while outputs at source lines to maximize the computing bandwidth.

(b) The 3D NAND device studied in this work

We measure a 16-layer 64Gb SLC SGVC 3D NAND chip to collect the I_{cell} distribution, as shown in **Fig. 5(a)**. Extremely-thin

body ($T_{si} \sim 3\text{nm}$, **Fig. 5(b)**) device is used to provide intrinsic lower mobility [7] to provide ultra-low $I_{on} \sim 2\text{nA}$, while keeping excellent ON/OFF ratio with extremely low I_{off} . **Figure 6(a)** shows a typical SLC distribution. With reduced $V_{pass,r} < 5\text{V}$ the SGVC device can possess very good read disturb immunity to pass $>10\text{G}$ read stress. The read disturb free property is very important for CIM because the computing is intensive during the life time. **Figure 6(b)** shows the I_{dVg} curves. The transistor is operated at the saturated region, where $V_g \gg V_t$ so that the I_{cell} is quite insensitive to the V_t shifts or noise. **Figure 6(c)** shows the I_{on} and I_{off} distributions. Mean $I_{on} \sim 2\text{nA}$, while I_{off} are all $< 0.5\text{pA}$. Thanks to the extremely-thin body device that provides excellent transistor characteristics. **Figure 6(d)** shows the collected I_{on} in the test chip. A full-block code (with 50% weight sparsity after training) is programmed and the array pattern loading effect is already considered.

(c) Design method to produce multi-bit resolution for MAC

Figure 7 illustrates an example for a 4-bit input and 4-bit weight (4I4W). 4bit resolution will soon become the mainstream edge computing. Although the device is SLC operation in both weight and input, we can use a “shifter and adder” design to produce 4bit. The 3 SSL’s corresponds to 2bit weight. We can add another block with 2bit shifter and adder to produce 4bit weight. For input, we use 3BL’s to represent 2bit input. Again, to duplicate the tile with 2bit shifter to enable 4bit input. With this method, we will distribute each weight in 36 different cells to represent 4I4W. Thanks to the very high-density 3D NAND so that we can largely duplicate the array multiple times without the need of resorting to difficult analog memory design.

The total BL number can be more than 10^4 (we typically make 13824 BL’s to support VGG network), still providing reasonable SA working range with low background leakage. The large parallelism is quite efficient to support the modern DNN network, where the input number often exceeds a few thousands.

(d) MAC currents and “Central Limit Theory”

By using the measured I_{cell} distribution in Fig. 6(d), we can randomly sum many cells to produce the MAC currents according to various numbers of summations. The MAC current distribution is amazingly tight as shown in **Fig. 8(a)**, even though the original I_{cell} distribution is quite wide. **Figure 8(b)** shows that the normalized MAC current is getting tighter as the string number increases. **Figure 9** shows that the normalized standard deviation of MAC is linearly proportional to the square root of N , where N is the string number for summation. It is consistent with the “Central Limit Theory”, which predicts that as the sample number increases, the mean of summation tends to be a normal distribution, and the normalized deviation shrinks ($1/\sqrt{N}$). Such a benefit comes from the merit of high parallelism of 3D NAND nvCIM to allow large number of summation from plural BL’s and SSL’s, thus the noise and variation will be averaged out to produce very tight MAC distribution centered around the mean value. This enables accurate MAC for higher resolution bit.

(e) Non-Ideal Cell Variation, Noise, and Shift

The major challenge of CIM is to deal with non-ideal cell distributions. In addition to the intrinsic I_{cell} distribution in Fig. 6(d), there are also noise (**Fig. 10**) and shifts (**Fig. 11**) that need to be considered to guarantee sufficient mass production window.

To provide accurate MAC, we need to develop adequate calibration technique for each SA, as shown in **Fig. 12**. Because training is done remotely, we can pre-know the target MAC value of each block. We can sacrifice one or two WL’s for each block to

store a known MAC value for calibration. To initialize the inference the SA needs to calibrate according to the known MAC value to trim the SA parameters to match the target value. Such calibration can be done “on-the-fly” to accommodate possible reliability aging effect. We expect the SA calibration of MAC to be the most critical design of CIM to ensure accuracy.

(f) Accuracy Simulation by Tensor Flow

We adopt the popular TensorFlow [8] simulator to simulate the VGG7 network with Cifar-10 database. The device noise and various errors are included in the simulator for CIM. The software trained network has accuracy of $\sim 90\%$ at 4bit resolution.

Figure 13 summarizes the accuracy simulation with various device error parameters. In **Fig. 13(a)**, the SA resolution requirement is 7~8 bit. The SA resolution requirement is more than 4bit, because the shifter and adder design together with large summation causes more resolution need. **Figure 13(b)** shows the accuracy can keep when I_{cell} sigma $< 0.5\text{nA}$ (measured $\sim 0.3\text{nA}$ in our chip). Thus we have good tolerance to I_{cell} variation, thanks to the central limit theory. **Figure 13(c)** shows the tolerance of negative shift is as high as -0.5nA (25%). However, **Fig. 13(d)** shows that the tolerance to the I_{cell} positive shift is only $+0.13\text{nA}$ ($\sim 6.5\%$). The reason is because the positive shift will make SA off-range. Thus calibration for positive shift is more critical.

(g) System simulation, and Summary

Figure 14 shows the MAC current distribution in VGG7. The average current is around $7\mu\text{A}$ for each SA, while maximum current is only $25\mu\text{A}$. It’s adequate for the low-power SA circuit design. Note that the natural high sparsity of inputs in neural network helps in reducing the power consumption. We forecast the TOPS/W is around 40 (4I4W), which is a few times better than LOGIC solutions ($\text{TOPS/W} < 10$). There are rooms to further improve the TOPS/W by sense amplifier design work. We suggest that the major showplace of CIM is for the high-density ($>100\text{Mb}$) neural network that is beyond the SRAM capability. In particular, CIM will find the best advantage of fully-connected (FC) network where weight is not re-used.

Figure 15 summarizes a predicted system performance to support a heavy network of VGG16, which is an extended version of VGG7. VGG16 requires large number of weight (138M in 4I4W) and huge MAC (15.5G in 4I4W). Such a big network is quite difficult to be supported by LOGIC solution with limited SRAM. In 3D NAND, we can duplicate cells many times to represent 4I4W, and the total memory density required is 6.5Gb for 3D NAND for an 16-core parallel operations. The forecasted speed is 13.4msec per frame, or 74.6 frame/sec, assuming minimum latency of data routing between adjacent layers for VGG16. The speed performance is already close to the advanced GPU solution, while the much cheaper transistor cost of 3D NAND still ensure low cost. Without the need of data movements from DRAM ($\sim \text{nJ/bit}$), 3D NAND nvCIM can greatly save the system power.

The VGG16 also generate sizable metadata of feature maps, which requires $\sim 3.2\text{MB}$ SRAM. It requires advanced CMOS + 3D NAND integration, which is challenging in the traditional way. Fortunately, the recent advancement of 3DIC such as tight-pitch Cu bond pad can enable system integration of 3D NAND with advanced CMOS to realize the 3D NAND nvCIM.

References:

- [1] M F Chang, VLSI 2019, Short Course [2] W H Chen et al, session31-4, pp 494-495, ISSCC 2018 [3] C X Xue et al, session24-1, pp 388-389, ISSCC 2019 [4] H T Lue, et al, pp 177-178, VLSI 2018 [5] <https://www.mythic-ai.com> [6] Panni Wang, et al, IEEE Tran VLSI, pp 988-991, 2019 [7] H T Lue, et al, pp 210-211, VLSI 2019 [8] <https://www.tensorflow.org>

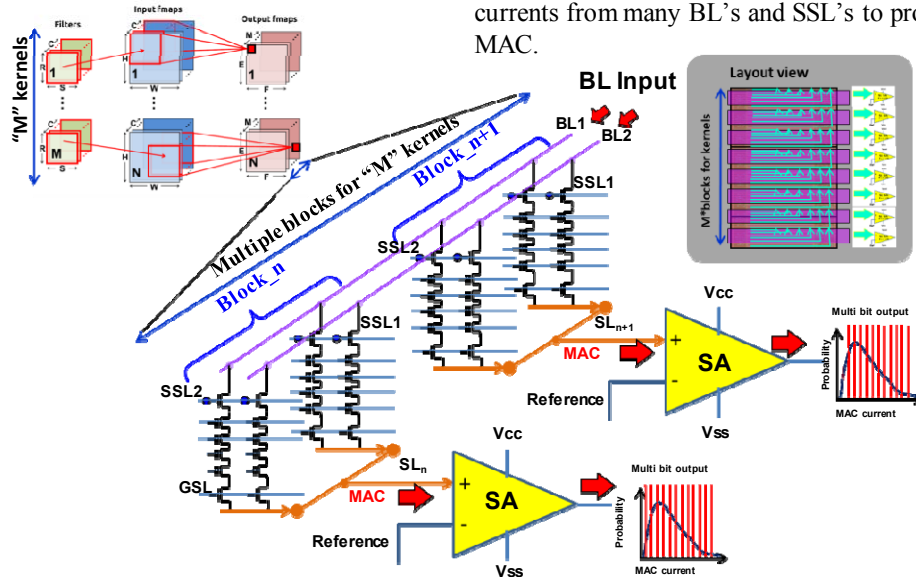
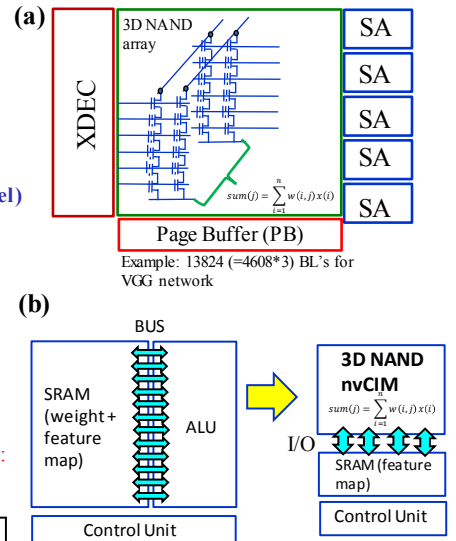
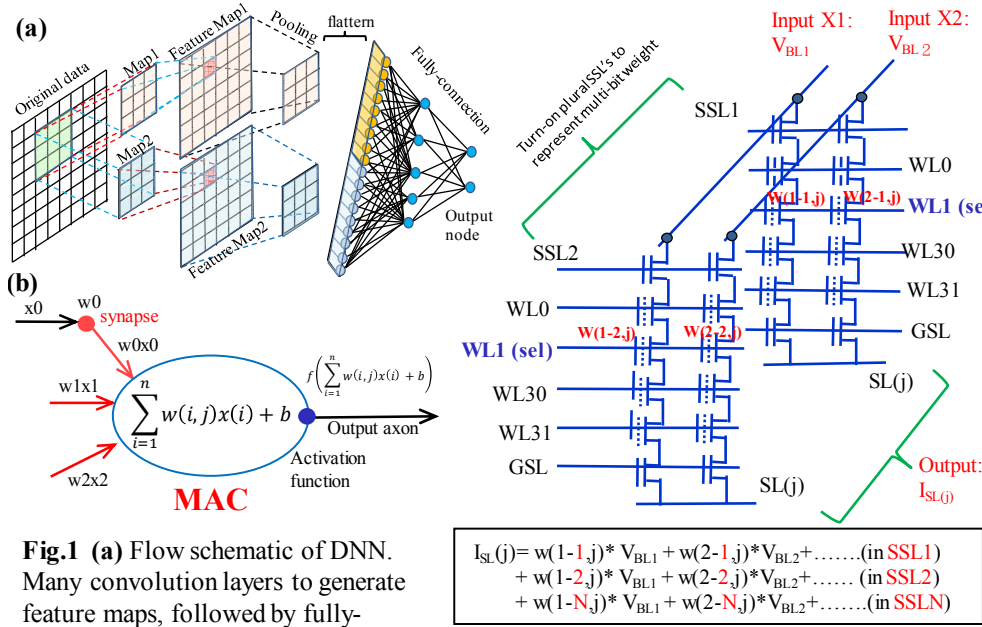
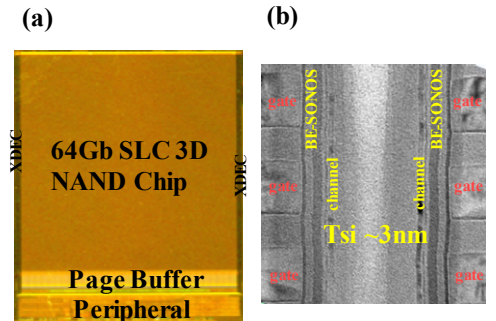
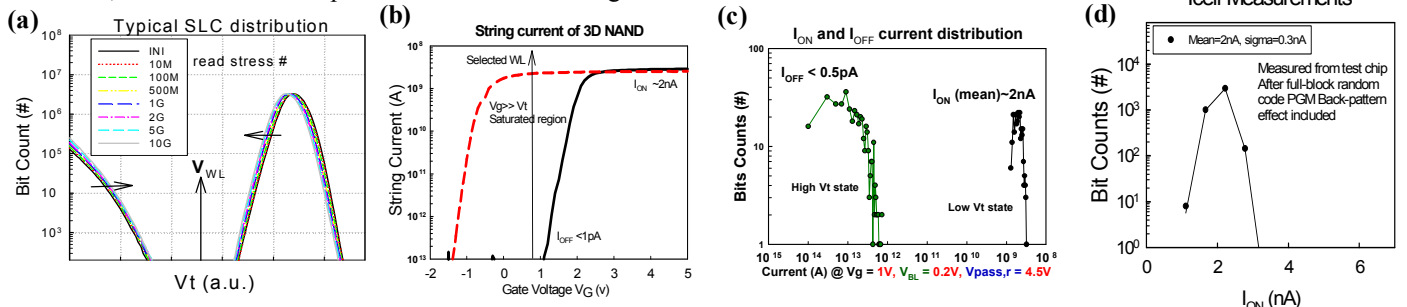


Fig.4 Overall design summary. The summed currents for each source line (SL) is connected to a multi-bit sense amplifier (SA) to digitize the MAC value. There are M blocks to mimic M kernels. BL input number depends on the input number of network, and SSL number depends on the multi-bit weight.



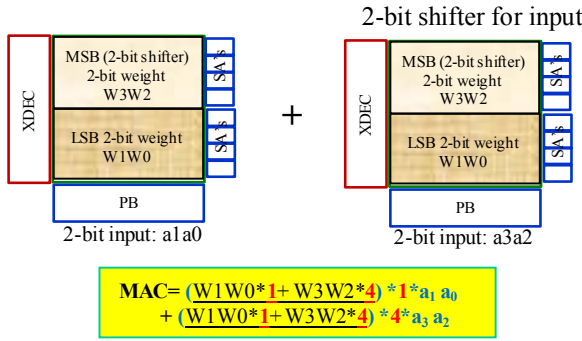


Fig.7 Methods to produce 4bit resolution. 3 SSL's to represent 2bit weight (W0W1). Another block and "shifter and adder" scheme to provide an effective 4bit weight. 3 BL's to represent 2bit input (a1a0) for each tile, while another tile and shifter and adder to represent 4bit input. Thus the each weight (SLC) is stored in 36 different locations to enable 4bit input and 4bit weight (4I4W).

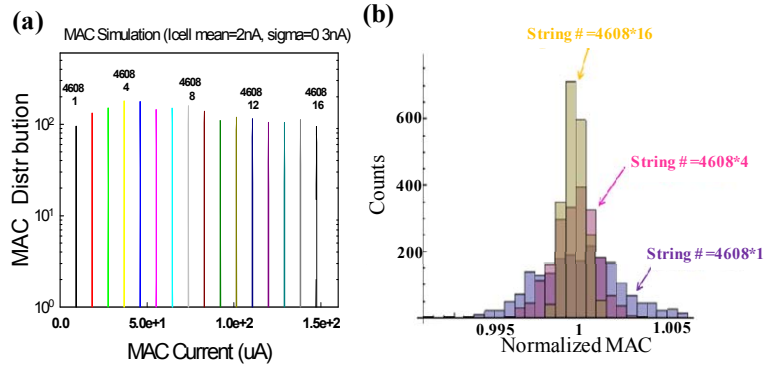


Fig.8 (a) MAC current simulation, assuming Icell follows Fig. 6(d). Various summed numbers are assumed. It shows very tight MAC distribution to support more than 4bit resolution. **(b)** The normalized MAC current distribution simulation. As the summed number is increased, the distribution is getting tighter.

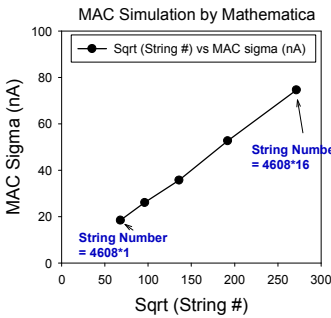


Fig.9 The standard deviation of MAC follows sqrt(N), where N is the number of strings. The normalized MAC $\propto 1/\sqrt{N}$.

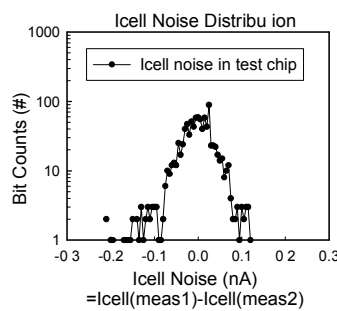


Fig.10 The typical Icell noise distribution in test chip, collected by Meas2-Mea1. Less than 0.1nA noise is found.

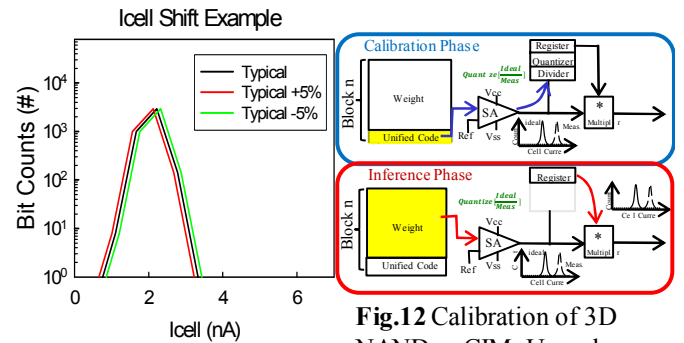


Fig.12 Calibration of 3D NAND nvCIM. Use a known code to calibrate the MAC value. Calibration needs more SA resolution.

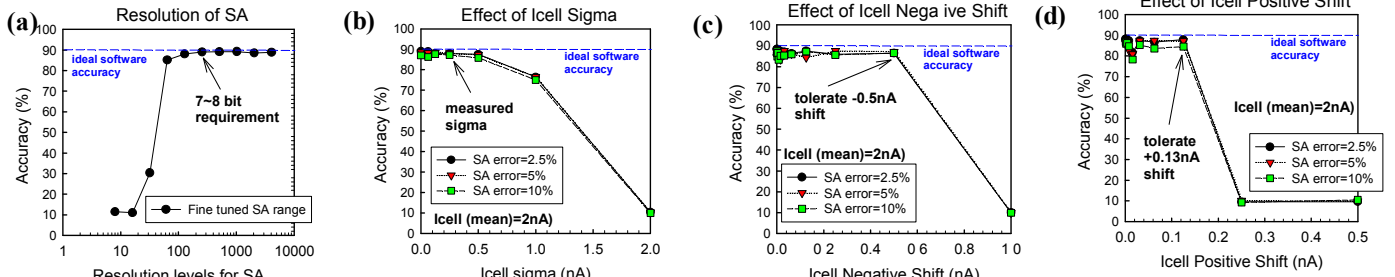


Fig.13 TensorFlow simulation results for 3D NAND nvCIM, considering various Icell variation and shifts error. SA error margin is also included. **(a)** The SA resolution may need 7-8bit to meet ~90% accuracy. **(b)** Tolerance of Icell variation sigma is 0.5nA for mean=2nA. **(c)** Tolerance of Icell mean negative shift is -0.5nA (25%). **(d)** Tolerance of Icell mean positive shift is +0.13nA (6.5%).

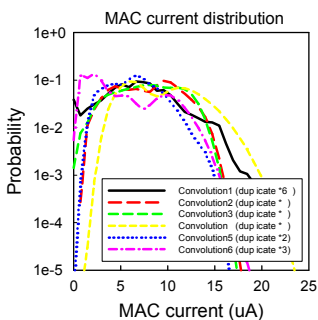


Fig.14 The MAC current distribution of 3D NAND nvCIM for a VGG7 network. MAC average current is around 7uA for each SA. Max. ~25uA.

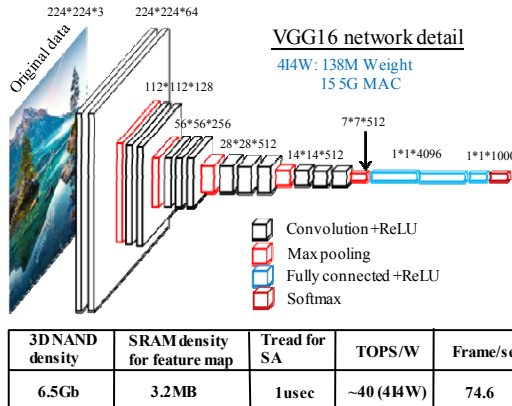


Fig.15 Summary of 3D NAND nvCIM for VGG16 computing (16 Cores assumption).

Convolution	BL # required for each tile	SA (block) # for each tile	# of Tread	Accumulated SA sensing time
Layer 1	81	64	3136	3 14 msec
Layer 2	1728	64	3136	3 14 msec
Layer 3	1728	128	1568	1 57 msec
Layer 4	3456	128	1568	1 57 msec
Layer 5	3456	256	784	784 usec
Layer 6	6912	256	784	784 usec
Layer 7	6912	256	784	784 usec
Layer 8	6912	512	392	392 usec
Layer 9	13824	512	392	392 usec
Layer 10	13824	512	392	392 usec
Layer 11	13824	512	98	98 usec
Layer 12	13824	512	98	98 usec
Layer 13	13824	512	98	98 usec
Fully-Connected				
Layer 14	75264	4096	64	64 usec
Layer 15	12288	4096	64	64 usec
Layer 16	12288	1000	16	16 usec
Total = 13.4 msec				